Rebecca De Rosa

Curriculum Vitae

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Summary

Cybersecurity graduate (MSc, Politecnico di Torino, 2026) working on LLM-based agents for software security. My Master's thesis built a multi-agent LLM system that augments CodeQL static analysis and detects substantially more vulnerabilities than CodeQL alone. As a research assistant at the University of Southern Denmark, I designed and evaluated an agentic vulnerability-detection pipeline with two industry partners, and I now lead the LLM development of a RAG-based architecture that supports university employees in their activities. I am interested in how to evaluate, verify and monitor the behaviour of LLM agents, and in building the benchmarks and methods that make their failures measurable.

Education

2024–2026 **MSc in Cybersecurity**, *Politecnico di Torino*, Turin, Italy

Graduated March 2026. Master's thesis carried out at the University of Southern Denmark (SDU), Sønderborg, as an international exchange student (2025–2026).

Thesis: *A Hybrid Multi-Agent Architecture for Enhancing CodeQL Static Analysis with Large Language Models*

Supervisors: Prof. Adam Alami (SDU), Prof. Danilo Giordano and Prof. Matteo Boffa (Politecnico di Torino)

Designed a three-agent LangGraph pipeline (Analyzer, Suggestor, Creator) that validates CodeQL results, diagnoses gaps in its rules, and drafts new CodeQL queries. On 26 real-world Python files from CVEfixes (CWE-78, CWE-79, CWE-89), the Analyzer agent identified the correct vulnerability in far more files than the CodeQL baseline. Compared GPT-5.2 and GPT-4o-mini under identical settings and evaluated generated rule proposals with an LLM-as-a-judge protocol; documented over-reporting and query syntax errors as limitations.

2020–2024 **BSc in Computer Engineering**, *Politecnico di Torino*, Turin, Italy

Preprints

2026 P. M. Konrad, T. L. Adam, A. C. H. Merrild, R. Terrenzi, **R. De Rosa**, T. Tanyel, S. Ayvaz. *The Open-Box Fallacy: Why AI Deployment Needs a Calibrated Verification Regime*. arXiv:2605.10601. [arXiv]

Argues that AI authorization should rest on scoped, independently checkable verification and post-deployment monitoring rather than on mechanistic explanation alone; proposes a reportable *Verification Coverage* metric.

Research Experience

Jul 2026– **Research Assistant, Applied and Interpretable Machine Learning Group**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark

Designed the workflow and implemented a retrieval-augmented generation (RAG) architecture that supports university employees in their daily activities. The source material is first processed with OCR (PaddleOCR) and converted into structured JSON files. The extracted content is split into chunks, using a chunking method adapted to the type of text (tables or paragraphs), and the chunks are turned into semantic embeddings stored in a vector database. Each user query is embedded and matched against the database, and an LLM generates the final answer from the retrieved content. I also built an interface where users can review the answers and see the source passages highlighted in the original.

Jan–Jun 2026 **Research Assistant**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark

Co-developed, in collaboration with two software companies (Umbraco and Unik), an AI system that analyses source code to find security vulnerabilities. LLM agents detect a vulnerability, explain how it could be exploited and propose a fix, which is tested automatically before a person reviews it.

Evaluated the system on a public vulnerability benchmark and on real open-source and production code, checking its findings against known vulnerabilities and with an independent LLM judge. It uncovered a likely exploitable flaw in an open-source library, and the evaluation showed how public vulnerability labels can mislead benchmark scores.

Teaching

Jul 2026– **Teaching Assistant, Web Development**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark

Bachelor-level course. Lead the exercise sessions, working through PHP and Laravel exercises with students.

2020–2024 **Private Tutor**, Turin, Italy

Physics, Mathematics, Chemistry and Computer Science for middle and high school students.

Technical Skills

LLMs & Agents	LangChain, LangGraph, single- and multi-agent design, closed and open-source models (GPT, Claude, DeepSeek, Qwen), design and implementation of RAG-based architectures, prompt engineering
ML	PyTorch
Security	Static analysis (CodeQL), vulnerability detection
Programming	Python, Java, C, C++, PHP (Laravel); Linux

Languages

Italian	Native
English	Advanced (C1)